

CIRO

User Flows & Personas

End-to-end journeys for every persona, role-based access matrix, and incident lifecycle

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This document describes the complete user journeys for all three CIRO personas: Citizen (reporting and tracking), Responder (field operations), and Admin/Dispatcher (triage, approval, and command). It includes step tables, decision points, the 14-state incident lifecycle, and the full role-based access control matrix.

Website	https://ciroquick.netlify.app
API Backend	https://bano-qabil-alibabacloud-hackthon-ciro-app-production.up.railway.app
Repository	https://github.com/saqib54/Bano-Qabil-AlibabaCloud-Hackthon-CIRO-APP

This document is part of the CIRO official documentation suite prepared for the Alibaba Cloud x Bano Qabil Hackathon. All content is based on the implemented system.

01

System Flow Overview

How citizens, AI, dispatchers, and responders interact

CIRO connects four actors in a linear pipeline with feedback loops. A citizen reports; the AI pipeline verifies; a human dispatcher approves; a responder acts in the field. Every stage pushes a real-time status update back to the citizen. Spam, duplicate, and low-confidence reports are diverted to human review queues before any real-world action is taken.

Actor	Entry Point	Key Action	Output
Citizen	Web / Android APK / iOS	Submit report (voice/photo/GPS)	Incident created, safety reply, live timeline
AI Pipeline	Automatic on submission	10-agent verification chain	Verdict + confidence + impact zone + copilot
Dispatcher	Admin portal AI Decision Center	Review verdict, approve/reject	Team dispatched or report rejected
Responder	Staff portal assignments	Accept, en-route, on-scene, resolved	Field update, resolution evidence
Citizen	My Reports timeline	Receive real-time WebSocket push	ETA, team details, status progress

02

Citizen: Report an Emergency — Happy Path

60-second flow from app launch to dispatched team

The citizen reporting flow is optimised for speed and accessibility. The entire process — from opening the app to receiving a safety reply — is designed to complete within 60 seconds even on a slow connection.

Step	User Action	Screen / Control	System Response
1	Open Report Emergency	Dashboard quick-action button	Location category selector displayed; voice and camera buttons visible
2	Choose input method	Category card OR voice button	Future: mic activates, live transcription shown in ur-PK locale
3	Speak / type description	Voice stream OR text field	AI autofill: category, location name, people count, priority extracted and populated
4	Capture photo (optional)	Camera button -> live viewfinder	User Media stream; citizen taps to capture; file uploaded as incident_media
5	Confirm GPS location	One-tap GPS button OR browser address bar	Browser Geolocation API fetches coordinates; GeoScout validates within PK
6	Add contact phone (optional)	Phone field	Used by SmartDispatch ETA calculation; stored in incident record
7	Submit report	Submit button	POST /incidents — incident INC-YYYYMMDD-#### created; 10-agent pipeline initiated
8	See instant safety reply	Success screen	Authority-approved safety template for the category displayed (< 1 s)
9	Monitor My Reports	My Reports screen	Live timeline updated by WebSocket: TRIAGING -> VERIFIED -> DISPATCHED
10	Receive dispatch notification	Push notification + My Reports screen	WebSocket: 'Team dispatched. ETA: X minutes. Department: [name]'

INFO Voice Recognition Fall-back: if ur-PK recognition fails or is unavailable, the system automatically retries with en-US locale. Roman Urdu typed input is also handled by the AI extract NLU layer.

03

Citizen: Passwordless Email OTP Sign-In

Secure 6-digit one-time code flow with rate limiting

Step	User Action	System Action	Notes
1	Enter email on sign-in page	POST /auth/otp/request	30-second cooldown enforced; dedicated rate limiter (10 req/min)
2a (Successful)	Check email inbox	6-digit code sent via SMTP (Resend button available)	Code is cryptographically hashed & stored in otp_codes table
2b (Demo mode)	See code on screen	devCode echoed in API response (devCode only)	NEVER returned in production NODE_ENV
3	Enter 6-digit code	POST /auth/otp/verify	10-minute expiry checked; attempt counter incremented
4 (Valid Code)	Redirected to dashboard	JWT access + refresh pair issued; provider_id, SUB, email, user record	provider_id, SUB, email, user record
4 (Invalid Code)	Error message shown	Attempt counter incremented; lock after 5 failed attempts	After 5 failed attempts, new code after lock
4 (Expired Code)	Code expired	otp_codes.expires_at checked; must not be past expiry	Code expired; user can re-request after cooldown

04

Dispatcher: AI-Assisted Triage & Approval

AI Decision Center queues, trace view, one-click dispatch

The dispatcher interacts with the AI Decision Center which surfaces three prioritised queues. For each incident, the dispatcher sees the full AI verdict card including confidence percentage, spam score, satellite signal, copilot summary, suggested team, and affected population estimate.

Queue	Trigger Condition	Recommended Action	Auto-Action Available
NEEDS_REVIEW	SpamGuard score >= 60 OR conflicting agent signals OR multiple copilot flags	Open incident card; verify against context; check agent signals	Yes (manual review required)
LOW_CONFIDENCE	VerdictEngine composite confidence < 0.55	Check satellite signal (typically 0.55+); request additional info or corroborate	Yes (request info)
DUPLICATE	DedupGuard found same category within 1km and 24hrs	Review linked incident; merge into existing case if required	Yes (merge)
AUTO_VERIFIED	Verdict = AUTO_VERIFIED AND Severity = HIGH AND Copilot = YES	System auto-dispatched; no manual intervention	Yes (pre-authorized)

Step	Dispatcher Action	System Response
1	Open AI Decision Center	Three queues loaded; incident cards with confidence badge, category, location, time
2	Click incident card	Expanded view: 10-agent per-stage trace, copilot summary, impact zone preview, suggested team
3	Review satellite signal	SUPPORTING / INCONCLUSIVE / CONFLICTING badge with source explanation
4a: Approve	Click 'Approve & Dispatch'	status -> DISPATCHED; dispatch_approved_by/at recorded; SmartDispatch assigns team; W
4b: Reject	Click 'Reject'	status -> FALSE_ALARM or REJECTED; audit_log: DISPATCH_REJECTED; citizen notified
5	Broadcast (optional)	Compose emergency broadcast targeted by area; fan-out to all connected citizens in range

05

Responder: Field Operations Flow

Verified, de-duplicated assignments with full field reporting

Step	Responder Action	System Response
1	Receive assignment notification	WebSocket incident.dispatched; push notification; assignment appears in dashboard
2	Open assignment detail	Full incident context: category, location pin, severity, people affected, copilot summary, AI-su
3	Accept assignment	PATCH /staff/assignments/:id/accept; status updated; dispatcher notified
4	Depart — set En Route	PATCH /:id/status {status: 'en_route'}; citizen WebSocket: 'Responder is on the way'
5	Arrive — set On Scene	PATCH /:id/status {status: 'on_scene'}; citizen WebSocket: 'Responder has arrived'
6	Add situation log	POST /assignments/:id/situation-log {note, optional photo}; visible to admin
7	Submit resolution	POST /assignments/:id/resolve {resolution_notes, proof_url, resources_used}; sent to admin
8	Admin approves resolution	PATCH /resolutions/:id/approve; status -> RESOLVED; citizen notified; incident closed

06

Role-Based Access Control Matrix

Who can do what — PUBLIC, STAFF, and ADMIN permissions

Capability / Action	PUBLIC (Citizen)	STAFF (Responder)	ADMIN (Dispatcher)
Register / Login / Google / OTP	Yes	Yes	Yes
Submit emergency report	Yes	Yes	Yes
View own reports + timeline	Yes	Yes	Yes
Cancel own report	Yes	Yes	Yes
Voice-to-Report / AI extract	Yes	Yes	Yes
Safety map (public incidents)	Yes	Yes	Yes
Emergency alerts / broadcasts	Yes	Yes	Yes
Weather forecast	Yes	Yes	Yes
AI Safety Assistant	Yes	Yes	Yes
Safe places / shelters (read)	Yes	Yes	Yes
Upload avatar / update profile	Yes	Yes	Yes
View assignment queue	No	Yes	Yes
Accept / update assignment status	No	Yes	Yes
Add situation log to incident	No	Yes	Yes

Capability / Action	PUBLIC (Citizen)	STAFF (Responder)	ADMIN (Dispatcher)
Submit resolution with evidence	No	Yes	Yes
Toggle duty status	No	Yes	Yes
View AI verdict trace per incident	No	No	Yes
Approve or reject AI verdict / dispatch	No	No	Yes
Issue emergency broadcast	No	No	Yes
Manage shelters (CRUD)	No	No	Yes
Manage users and staff	No	No	Yes
Manage departments	No	No	Yes
View audit logs	No	No	Yes
Approve / reject resolutions	No	No	Yes
View analytics dashboard	No	No	Yes
Access forecast hotspots	No	No	Yes
Modify system settings	No	No	Yes

07

Incident Status Lifecycle — 14 States

Enforced transition rules from submission to closure

State	Triggered By	Valid Next States	Notes
REPORTED	Citizen submits	TRIAGING, CANCELLED	INC number assigned; AI pipeline queued
TRIAGING	AI pipeline starts (async)	VERIFIED, NEEDS_REVIEW, REJECTED, DUPLICATE	AI pipeline outputs, incident stored
AI_ANALYZED	Pipeline complete (internal)	VERIFIED, NEEDS_REVIEW	Internal transition before dispatcher sees it
VERIFIED	AUTO_VERIFIED (high confidence)	DISPATCHED	Auto-alert may be issued; dispatcher can still override
NEEDS_REVIEW	SpamGuard flagged or low confidence	VERIFIED, REJECTED	Human dispatcher decision required
DUPLICATE	DedupGuard match	VERIFIED (if independent), REJECTED	Linked to master incident
FALSE_ALARM	Dispatcher rejects	(terminal)	Audit-logged; citizen notified
REJECTED	Dispatcher rejects / spans out	(terminal)	Audit-logged; no team dispatched
DISPATCHED	Dispatcher approves	EN_ROUTE	Team assigned; citizen gets ETA
ASSIGNED	SmartDispatch assigns team	RESPONDING	Responder accepts
RESPONDING	Responder accepts and en route	ON_SCENE	Citizens gets real-time status
ON_SCENE	Responder arrives	RESOLVED	Situation logs can be added
RESOLVED	Evidence submitted by responder	CLOSED, REOPENED	Admin approves resolution with evidence
CLOSED	Admin approves resolution	(terminal)	Before/after evidence stored
CANCELLED	Citizen cancels (REPORTED state)	(terminal)	Only in REPORTED state; citizen can cancel
REOPENED	Admin reopens closed incident	RESPONDING	For follow-up or incomplete resolution